

Bio Informatics

Lecture 18

Profile Alignments

Scoring matrices discussed so far are used in pairwise sequence alignment (previous class) .

Can be used to;

- estimate the evolutionary distance between a pair of proteins.
- In a sequence data base, find proteins similar to a given proteins (we will discuss methods to do so in the next class).

A different scenario: we have a set of related proteins and ask the question if a given sequence is a member of this set.

Profile Alignments - Goals

- Given: Sample sequences of a protein family.
- Goal: provide a stochastic model of the family.
- Method:
 - Multiple align sequences in the family.
 - Construct position specific score matrix: the score $s_i(a)$ of amino acid a on the position i and depends on frequency of observing a in this position in multiple alignment.
 - Usual considerations – should sequences in the family be weighted equally?
 - Pseudo-counter – create small frequencies for amino-acids not represented in the profile.

Profile Alignment – PWM or PSSM

- A **position weight matrix (PWM)**, also known as a **position-specific weight matrix (PSWM)** or **position-specific scoring matrix (PSSM)**, is a commonly used representation of [motifs](#) (patterns) in biological sequences.
- PWMs are often derived from a set of aligned sequences that are thought to be functionally related and have become an important part of many software tools for computational motif discovery.

Profile Alignment – PWM or PSSM

- A PWM has one row for each symbol of the alphabet (4 rows for nucleotides in DNA sequences or 20 rows for amino acids in protein sequences) and one column for each position in the pattern.
- In the first step in constructing a PWM, a basic position frequency matrix (PFM) is created by counting the occurrences of each nucleotide at each position.

Profile Alignment – PWM or PSSM

- From the PFM, a position probability matrix (PPM) can now be created by dividing that former nucleotide count at each position by the number of sequences, thereby normalizing the values.
- Formally, given a set X of N aligned sequences of length L , the elements of the PPM $\mathbf{M}$ are calculated:

Profile Alignment – PWM or PSSM

- Formally, given a set X of N aligned sequences of length l , the elements of the PPM $\mathbf{M}$ are calculated:

$$M_{k,j} = \frac{1}{N} \sum_{i=1}^N I(X_{i,j} = k),$$

where $i \in (1, \dots, N)$, $j \in (1, \dots, l)$, k is the set of symbols in the alphabet and $I(a=k)$ is an indicator function where $I(a=k)$ is 1 if $a=k$ and 0 otherwise.

PSSM - Example

- For example we have a family of 10 DNA Sequences.

1. GAGGTAAAC
2. TCCGTAAGT
3. CAGGTTGGA
4. ACAGTCAGT
5. TAGGTCATT
6. TAGGTACTG
7. ATGGTAACT
8. CAGGTATAC
9. TGTGTGAGT
10. AAGGTAAGT

PFM – Position Frequency Matrix (Step-1)

A	3								
C	2								
G	1								
T	4								

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PFM – Position Frequency Matrix (Step-1)

A	3	6	1	0	0	6	7	2	1
C	2	2	1	0	0	2	1	1	2
G	1	1	7	10	0	1	1	5	1
T	4	1	1	0	10	1	1	2	6

1. GAGGTAAAC
2. TCCGTAAGT
3. CAGGTTGGA
4. ACAGTCAGT
5. TAGGTCATT
6. TAGGTACTG
7. ATGGTAACT
8. CAGGTATAC
9. TGTGTGAGT
10. AAGGTAAGT

PPM – Position Probability Matrix (Step-2)

A	3	6	1	0	0	6	7	2	1
C	2	2	1	0	0	2	1	1	2
G	1	1	7	10	0	1	1	5	1
T	4	1	1	0	10	1	1	2	6

A	0.3								
C	0.2								
G	0.1								
T	0.4								

PPM – Position Probability Matrix (Step-2)

A	0.3	0.6	0.1	0.0	0.0	0.6	0.7	0.2	0.1
C	0.2	0.2	0.1	0.0	0.0	0.2	0.1	0.1	0.2
G	0.1	0.1	0.7	1.0	0.0	0.1	0.1	0.5	0.1
T	0.4	0.1	0.1	0.0	1.0	0.1	0.1	0.2	0.6

For example, the probability of the sequence $S = \text{GAGGTAAAC}$ given the above PPM $\mathbf{M}$ can be calculated:

$$p(S|M) = 0.1 \times 0.6 \times 0.7 \times 1.0 \times 1.0 \times 0.6 \times 0.7 \times 0.2 \times 0.2 = 0.0007056.$$

Conversion From PPM to PWM – (Step-3)

- Most often the elements in PWMs are calculated as log likelihoods. That is, the elements of a PPM are transformed using a background model b so that:

$$M_{k,j} = \log_2 (M_{k,j} / b_k)$$

That is, the value of $b_k = 1/|k|$ for all symbols in the alphabet
(0.25 for nucleotides and 0.05 for amino acids)

Conversion From PPM to PWM – (Step-3)

A									
C									
G									
T									

$$M_{k,j} = \log_2 (M_{k,j} / b_k)$$

That is, the value of $b_k = 1/|k|$ for all symbols in the alphabet
(0.25 for nucleotides and 0.05 for amino acids)

Conversion From PPM to PWM – (Step-3)

$$M = \begin{matrix} A \\ C \\ G \\ T \end{matrix} \begin{bmatrix} 0.26 & 1.26 & -1.32 & -\infty & -\infty & 1.26 & 1.49 & -0.32 & -1.32 \\ -0.32 & -0.32 & -1.32 & -\infty & -\infty & -0.32 & -1.32 & -1.32 & -0.32 \\ -1.32 & -1.32 & 1.49 & 2.0 & -\infty & -1.32 & -1.32 & 1.0 & -1.32 \\ 0.68 & -1.32 & -1.32 & -\infty & 2.0 & -1.32 & -1.32 & -0.32 & 1.26 \end{bmatrix}$$

- When the PWM elements are calculated using log likelihoods, the score of a sequence can be calculated by adding (rather than multiplying) the relevant values at each position in the PWM.
- The score is 0 if the sequence has the same probability of being a functional site and of being a random site.
- The score is greater than 0 if it is more likely to be a functional site than a random site, and less than 0 if it is more likely to be a random site than a functional site.

PSSM or PWM - Uses

- There are various algorithms to scan for hits of PWMs in sequences.
- One example is the MATCH algorithm which has been implemented in the ModuleMaster.
- More sophisticated algorithms for fast database searching with nucleotide as well as amino acid PWMs/PSSMs are implemented in the possumsearch software.